

Math Thermo

class # 02

01/22/2020

Let us recall the example from the last lecture.

Example (1.8): Consider a unary ($r=1$) isolated system in equilibrium with the fundamental relation

$$\tilde{S}(U, V, N) = \left(\frac{N V U R^2}{v_0 \theta} \right)^{1/3}.$$

We can set

$$\Sigma_v = [0, \infty)^3.$$

$\tilde{S} = 0$ for all $(U, V, N) \in \partial \Sigma_v$, where

$$\partial \Sigma_v = \{(U, V, N) \in \Sigma_v \mid U=0, \text{ or } V=0, \text{ or } N=0\}.$$

Recall,

$$\begin{aligned} \frac{1}{\frac{\partial \tilde{S}}{\partial U}} &= T_S(U, V, N) = \frac{3 v_0 \theta}{N V R^2} \left(\tilde{S}(U, V, N) \right)^2 \\ &= \frac{3 v_0 \theta}{N V R} \left(\frac{N V U R^2}{v_0 \theta} \right)^{2/3}. \end{aligned}$$

T_S is not defined for all of Σ_v ! It is defined for all

$$(U, V, N) \in \text{dom}(T_S) \overset{\text{proper}}{\subset} \Sigma_v$$

Define

$$A_Z := \{(U, V, N) \in \text{dom}(T_S) \mid T_S(U, V, N) = 0\}.$$

AZ is called the absolute zero set.

Then

$$\overset{\text{proper}}{\downarrow} \quad AZ \subset \partial \Sigma_U. \quad (\text{Postulate V})$$

I leave it as a homework exercise to determine $\text{dom}(T_S)$ and AZ.

Remark: From this point forward, we will assume that all postulates hold. (We may one or two more later.)

Theorem (2.1) Suppose that in a composite system α and β are separated by a diathermal wall. Then, the equilibrium of the composite system may be characterized by

$$(2.1) \quad U^\alpha + U^\beta = U_0$$

and

$$(2.2) \quad T_S^\alpha(U^\alpha, V^\alpha, \vec{N}^\alpha) = T_S^\beta(U^\beta, V^\beta, \vec{N}^\beta).$$

Proof: Internal energy may be exchanged between systems α and β but cannot be exchanged with the outside world. Thus (2.1) must hold because of energy conservation. At equilibrium we must have

$$S(U^\alpha) = S^\alpha(U^\alpha, V^\alpha, \vec{N}^\alpha) + S^\beta(U_0 - U^\alpha, V^\beta, \vec{N}^\beta)$$

and

$$\frac{\partial S}{\partial U^\alpha} = 0.$$

Note, all other variables besides U^α are fixed.

$$\begin{aligned}
 0 &= \frac{\partial S}{\partial U^\alpha} = \frac{\partial S^\alpha}{\partial U^\alpha} + \frac{\partial S^\beta}{\partial U^\beta} \frac{\partial}{\partial U^\alpha} (U_0 - U^\alpha) \\
 &= \left[T_s^\alpha(U^\alpha, V^\alpha, \vec{N}^\alpha) \right] + \left[T_s^\beta(U_0 - U^\alpha, V^\beta, \vec{N}^\beta) \right] (-1)
 \end{aligned}$$

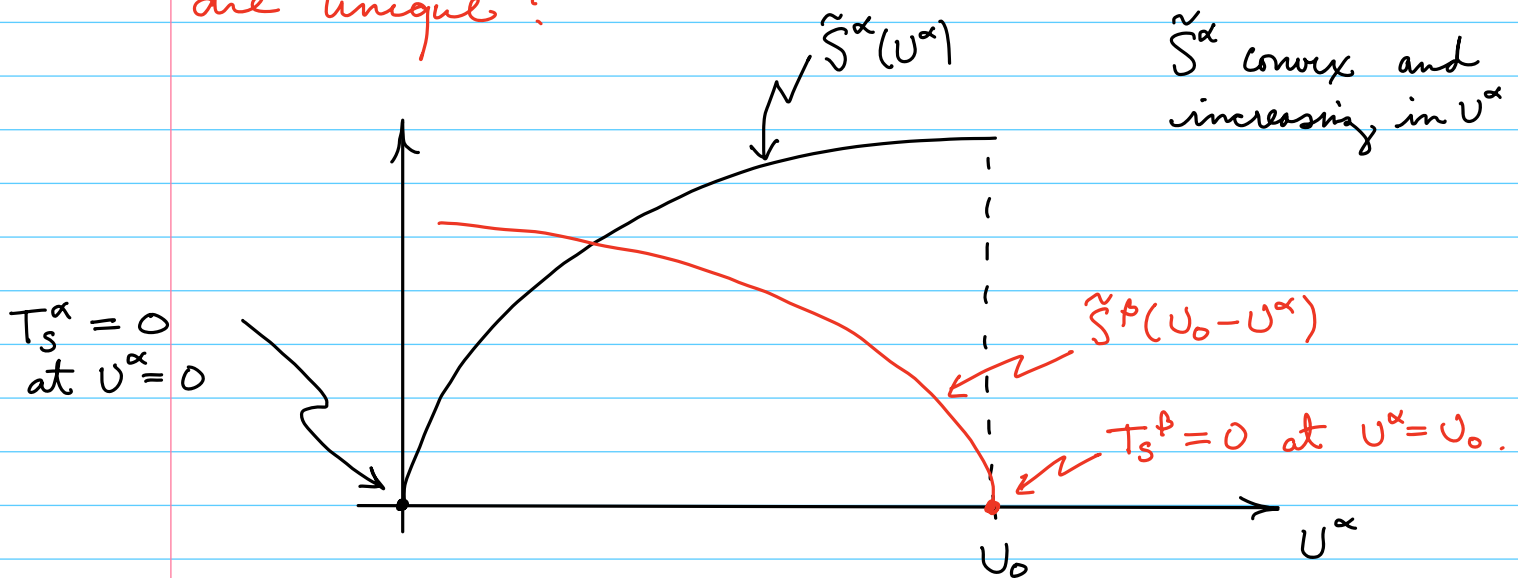
Thus,

$$T_s^\alpha(U^\alpha, V^\alpha, \vec{N}^\alpha) = T_s^\beta(U_0 - U^\alpha, V^\beta, \vec{N}^\beta)$$

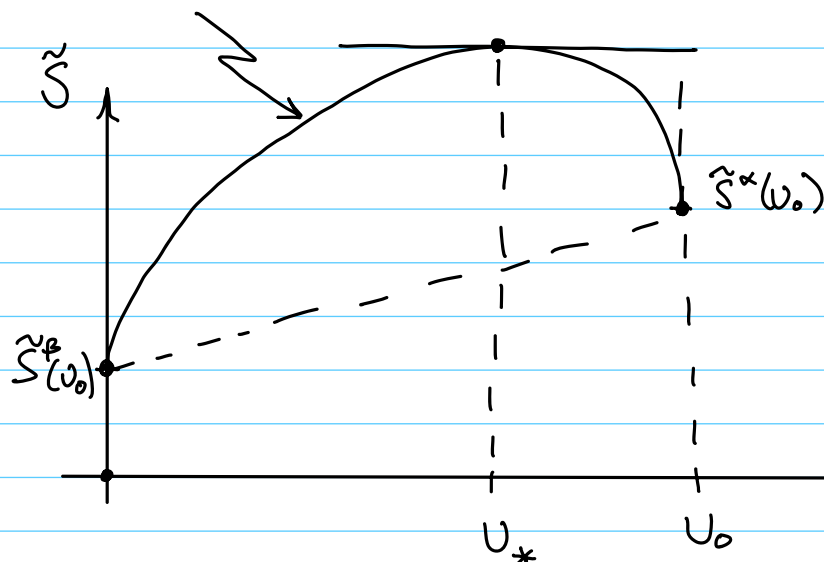
with

$$U^\beta = U_0 - U^\alpha$$

How do we know that solutions exist and are unique?



$$\tilde{S}(U) = \tilde{S}^\alpha(U) + \tilde{S}^\beta(U_0 - U)$$



We can turn what we know from the postulates into a proof that there is a unique max.

This proof can also be carried out by using Lagrange multipliers. Set

$$J := S^\alpha(U^\alpha, V^\alpha, \vec{N}^\alpha) + S^\beta(U^\beta, V^\beta, \vec{N}^\beta) + \lambda(U^\alpha + U^\beta - U_0)$$

Then, equilibrium is characterized by

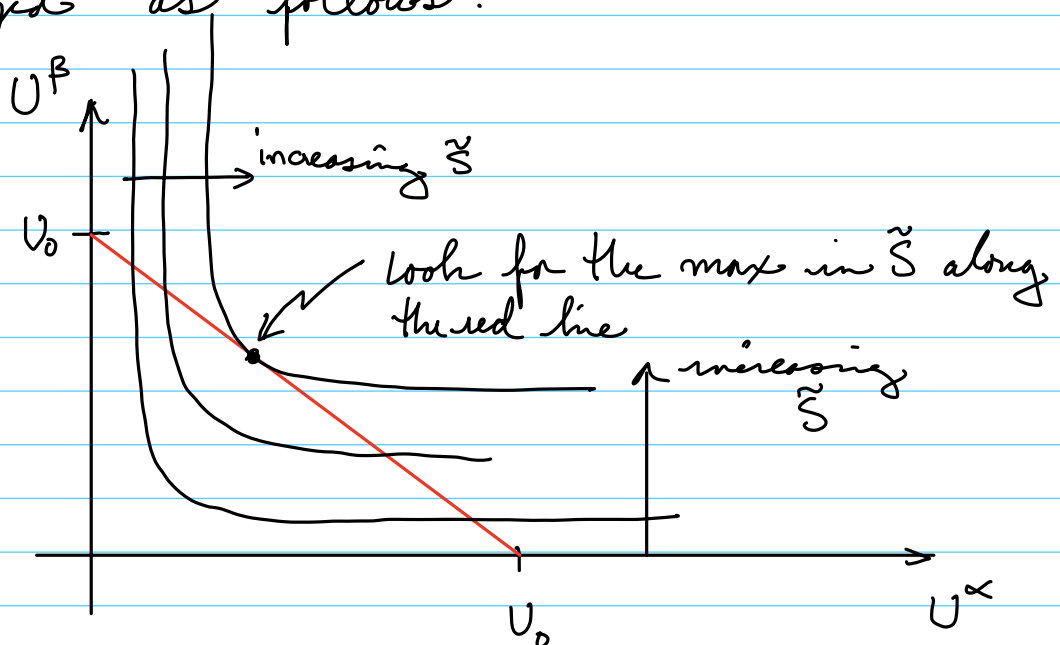
$$0 = \frac{\partial J}{\partial U^\alpha} = [T_{\text{S}}^\alpha(U^\alpha, V^\alpha, \vec{N}^\alpha)]^{-1} + \lambda,$$

$$0 = \frac{\partial J}{\partial U^\beta} = [T_{\text{S}}^\beta(U^\beta, V^\beta, \vec{N}^\beta)]^{-1} + \lambda,$$

$$0 = \frac{\partial J}{\partial \lambda} = U^\alpha + U^\beta - U_0,$$

which yields the same result. ///

The Lagrange Multiplier technique can be visualized as follows:



Here $\tilde{S}$ is a function of U^α and U^β .

Theorem (2.2) Suppose that in a composite system α and β are separated by a diathermal piston. Then, the equilibrium of the composite system may be characterized by

$$(2.3) \quad U^\alpha + U^\beta = U_0,$$

$$(2.4) \quad V^\alpha + V^\beta = V_0,$$

and

$$(2.5) \quad T_S^\alpha(U^\alpha, V^\alpha, \vec{N}^\alpha) = T_S^\beta(U^\beta, V^\beta, \vec{N}^\beta), \quad (\text{thermal equil.})$$

$$(2.6) \quad P_S^\alpha(U^\alpha, V^\alpha, \vec{N}^\alpha) = P_S^\beta(U^\beta, V^\beta, \vec{N}^\beta). \quad (\text{mechanical equil.})$$

Proof: For this let us use the method of Lagrange multipliers. Note that $\vec{N}^\alpha$ and $\vec{N}^\beta$ are fixed. Define

$$\begin{aligned} J := & S^\alpha(U^\alpha, V^\alpha, \vec{N}^\alpha) + S^\beta(U^\beta, V^\beta, \vec{N}^\beta) \\ & + \lambda_U (U^\alpha + U^\beta - U_0) \\ & + \lambda_V (V^\alpha + V^\beta - V_0). \end{aligned}$$

The conditions for equilibrium are

$$0 = \frac{\partial J}{\partial U^\alpha} = \left[T_S^\alpha(U^\alpha, V^\alpha, \vec{N}^\alpha) \right]^{-1} + \lambda_U,$$

$$0 = \frac{\partial J}{\partial U^\beta} = \left[T_S^\beta(U^\beta, V^\beta, \vec{N}^\beta) \right]^{-1} + \lambda_U,$$

$$0 = \frac{\partial J}{\partial V^\alpha} = \frac{-P_S^\alpha(U^\alpha, V^\alpha, \vec{N}^\alpha)}{T_S^\alpha(U^\alpha, V^\alpha, \vec{N}^\alpha)} + \lambda_V,$$

$$0 = \frac{\partial J}{\partial V^\beta} = \frac{-P_S^\beta(U^\beta, V^\beta, \vec{N}^\beta)}{T_S^\beta(U^\beta, V^\beta, \vec{N}^\beta)} + \lambda_V.$$

and

$$0 = \frac{\partial J}{\partial \lambda_U} = U^\alpha + U^\beta - U_0,$$

$$0 = \frac{\partial J}{\partial \lambda_V} = V^\alpha + V^\beta - V_0.$$

The result is clear. ///

Theorem (2.3): Suppose that a composite system is composed of two otherwise isolated systems with no barrier between the systems/phases. Suppose that

$$S^\alpha = S^\alpha(U^\alpha, V^\alpha, \vec{N}^\alpha)$$

and

$$S^\beta = S^\beta(U^\beta, V^\beta, \vec{N}^\beta)$$

are the fundamental entropy relations for the two systems. Then the equilibrium state is defined by the relations

$$(2.7) \quad U^\alpha + U^\beta = U_0$$

$$(2.8) \quad V^\alpha + V^\beta = V_0$$

and

$$(2.9) \quad N_i^\alpha + N_i^\beta = N_{0,i}$$

where $U_0, V_0, N_{0,i} > 0$, and

$$(2.10) \quad T^\alpha(U^\alpha, V^\alpha, \vec{N}^\alpha) = T^\beta(U^\beta, V^\beta, \vec{N}^\beta) \quad (\text{thermal equil.})$$

$$(2.11) \quad P^\alpha(U^\alpha, V^\alpha, \vec{N}^\alpha) = P^\beta(U^\beta, V^\beta, \vec{N}^\beta) \quad (\text{mech. equil.})$$

$$(2.12) \quad \mu_i^\alpha(U^\alpha, V^\alpha, \vec{N}^\alpha) = \mu_i^\beta(U^\beta, V^\beta, \vec{N}^\beta) \quad (\text{chem. equil.})$$

Proof: The procedure is the same. One can use the method of Lagrange multipliers to do the calculation. In particular, define

$$J := S^\alpha(U^\alpha, V^\alpha, \vec{N}^\alpha) + S^\beta(U^\beta, V^\beta, \vec{N}^\beta) + \lambda_0 (U^\alpha + U^\beta - U^0) \\ + \lambda_v (V^\alpha + V^\beta - V_0) + \sum_{i=1}^r \lambda_i (N_i^\alpha + N_i^\beta - N_{0,i}). \quad //$$

Example (2.4): Suppose that, for an isolated many fluid

$$\tilde{U} = \left(\frac{v_0 \theta}{R^2} \right) \frac{S^3}{NV}, \quad \Sigma_s \subset [0, \infty)^3.$$

A explicit expression of the form $\tilde{U} = \tilde{U}(S, V, N)$ is called a fundamental relation.

Here v_0 , θ , and R are positive constants.

The units of U and $\tilde{U}$ are

$$[U] = \text{Joules}.$$

The units of entropy, S and $\tilde{S}$, are

$$[S] = \frac{\text{Joules}}{\text{degree Kelvin}} = \frac{J}{K}$$

The units of volume, V , are

$$[V] = \text{meters}^3 \text{ or liters}$$

The units of N are

$$[N] = \text{moles}$$

Of course, it is easy to see that

$$\tilde{S} = \left(\frac{NVUR^2}{v_0\theta} \right)^{1/3}, \quad \Sigma_v = [0, \infty)^3.$$

An explicit function of the form

$$\tilde{S} = \tilde{S}(v, V, N)$$

is also called a fundamental relation

In any case, the temperature which has units

$$[T] = \text{degrees Kelvin} = K,$$

is

$$T_v(S, V, N) = \frac{\partial \tilde{U}}{\partial S} = \frac{3v_0\theta}{R^2} \frac{S^2}{NV}$$

This expression is called an equation of state.

Observe that

$$T_v(\lambda S, \lambda V, \lambda N) = T_v(S, V, N),$$

that is T_v is homogeneous of order 0.

This property is true of every equation of state, as we shall see.

Now,

$$\begin{aligned}\left(\frac{\partial \tilde{S}}{\partial U}\right)^{-1} &= \left[\frac{1}{3} \left(\frac{NVUR^2}{v_0 \theta} \right)^{-2/3} \frac{NVUR^2}{v_0 \theta} \right]^{-1} \\ &= \frac{3v_0 \theta}{NVUR^2} \left(\frac{NVUR^2}{v_0 \theta} \right)^{2/3} \\ &= \frac{3v_0 \theta}{NVUR^2} \left(\tilde{S}(U, V, N) \right)^2 \\ &= T_S(U, V, N).\end{aligned}$$

Clearly

$$T_U(\tilde{S}(U, V, N), V, N) = T_S(U, V, N),$$

as claimed in Theorem (1.7).

The pressure satisfies the equation of state

$$\begin{aligned}P_U(S, V, N) &= -\frac{v_0 \theta}{R^2} \frac{S^3}{N} \frac{-1}{V^2} \\ &= \frac{v_0 \theta}{R^2} \frac{S^3}{NV^2},\end{aligned}$$

which is also homogeneous of order zero.

We leave it as an exercise for the reader to show that

$$P_S(U, V, N) = P_U(\tilde{S}(U, V, N), V, N).$$

Finally, the chemical potential is

$$\mu_v(S, V, N) = -\frac{v_0 \theta}{R^2} \frac{S^3}{V N^2},$$

which is clearly homogeneous of degree zero.

The reader can show that

$$\mu_s(U, V, N) = \mu_v(\tilde{S}(U, V, N), V, N).$$

Recall, we have assumed with Postulate II that S is homogeneous of degree one. We also must have the following, as suggested by the example.

Theorem (2.5): $\tilde{U}$ is homogeneous of degree one when written as a function of $S, V, \vec{N}$:

$$\tilde{U} = \tilde{U}(S, V, \vec{N}).$$

Further, $T_v(S, V, \vec{N})$, $P_v(S, V, \vec{N})$, and $\mu_{v,i}(S, V, \vec{N})$, the equations of state, are homogeneous of degree zero, meaning

$$T_v(\lambda S, \lambda V, \lambda \vec{N}) = T_v(S, V, \vec{N}),$$

for any $\lambda > 0$, and similarly for P and $\mu_{v,i}$, $i=1, \dots, r$.

likewise $T_s(U, V, \vec{N})$, $P_s(U, V, \vec{N})$, and $\mu_{s,i}(U, V, \vec{N})$ are homogeneous of degree zero.

Proof: Fix $V \in [0, \infty)$ and $\vec{N} \in [0, \infty)^r$. $\tilde{S}$ is a monotonically increasing function of $U \in [0, \infty)$. For each $S \in [0, \infty)$ there exists a unique $U \in [0, \infty)$ such that

$$(2.13) \quad S = \tilde{S}(U, V, \vec{N}),$$

where we assume, for simplicity, that $\Sigma_U = [0, \infty)^{r+2}$.

$$(2.14) \quad \text{Then, } \tilde{U}(S, V, \vec{N}) = \tilde{U}(\tilde{S}(U, V, \vec{N}), V, \vec{N}).$$

let $\lambda > 0$ be arbitrary. Then,

$$(2.15) \quad \tilde{U}(\lambda S, \lambda V, \lambda \vec{N}) = \tilde{U}(\lambda \tilde{S}(U, V, \vec{N}), \lambda V, \lambda \vec{N}).$$

Since $\tilde{S}$ is homogeneous of degree 1, it follows that

$$(2.16) \quad \lambda \tilde{S}(U, V, \vec{N}) = \tilde{S}(\lambda U, \lambda V, \lambda \vec{N}).$$

Also, recall that, generically,

$$(2.17) \quad \tilde{U}(\tilde{S}(\hat{U}, \hat{V}, \hat{\vec{N}}), \hat{V}, \hat{\vec{N}}) = \hat{U}$$

because of inverse relations. We have

$$\begin{aligned} \tilde{U}(\lambda S, \lambda V, \lambda \vec{N}) &\stackrel{(2.15)}{=} \tilde{U}(\lambda \tilde{S}(U, V, \vec{N}), \lambda V, \lambda \vec{N}) \\ &\stackrel{(2.16)}{=} \tilde{U}(\tilde{S}(\lambda U, \lambda V, \lambda \vec{N}), \lambda V, \lambda \vec{N}) \\ &\stackrel{(2.17)}{=} \lambda U \\ &\stackrel{(2.17)}{=} \lambda \tilde{U}(\tilde{S}(U, V, \vec{N}), V, \vec{N}) \\ &\stackrel{(2.13)}{=} \lambda \tilde{U}(S, V, \vec{N}). \end{aligned}$$

Since, $S, V, \vec{N}$ are arbitrary, the result is proved.

We leave the other demonstrations for exercises. |||